

Care Access Navigator

Rubric Criterion 3: Implement Threat Intelligence Principles | HealthLine Regional Medical Center (HRMC)

Indicators of Compromise (IOC) Analysis

IN PLAIN TERMS

Two clear warning signs (“indicators”) came out of the malware analysis: the exact digital fingerprint of the malicious file, and the address it tried to send stolen data to. Either one showing up on HRMC's network would be a strong, reliable signal that this specific threat is present, not a false alarm.

IOC	Type	Detection Method	How It Indicates a Threat
1c72369e2543722e89be3a14448d0fdc1664f4638d91d67720dddb7ff73cb736	File hash (SHA256)	Hash-based lookup against threat intelligence sources (VirusTotal); 34/70 vendor engines flagged this exact file as malicious	A hash is a unique fingerprint of one specific file. A match confirms with near-certainty that this exact known-malicious executable (AgentTesla) is present on a host, not just something similar
smtp.hostinger.com	Network indicator (domain)	Network/DNS traffic monitoring; observed via sandbox detonation as the destination for outbound SMTP traffic on port 587	This domain is not a known HRMC-approved mail relay. Any internal host contacting it over SMTP is a strong signal of credential exfiltration in progress, independent of which specific malware variant is responsible

Application at HRMC: Both indicators would be added to endpoint/EDR and network monitoring blocklists. The file hash supports host-based detection (identifying infected machines), while the domain indicator supports network-based detection (catching exfiltration attempts even from variants or repackaged versions of the malware that produce a different file hash).

OpenCTI Threat Intelligence Platform

IN PLAIN TERMS

OpenCTI is a free tool that organizes threat information (malware, attacker groups, IOCs, and the connections between them) into one searchable, linked database, instead of scattered across separate documents. This section explains what it is, whether Care Access Navigator actually needs it, and how it would connect to the work already documented for this project. Per instructor guidance, only documentation is required at this stage — live deployment is proposed as future work, not yet built.

What OpenCTI Is

OpenCTI (Open Cyber Threat Intelligence) is an open-source platform for structuring and correlating threat intelligence data. It represents malware, threat actors, campaigns, indicators of compromise (file hashes, domains, IPs), and the relationships between them using the STIX2 industry standard, creating a searchable knowledge graph rather than a flat list of facts. It can automatically ingest external threat feeds through **connectors** — small automated importers,

such as one that loads the entire MITRE ATT&CK; framework or one that pulls community-submitted IOCs from a feed like AlienVault OTX.

Is OpenCTI Necessary for Care Access Navigator?

Question	Finding
Is OpenCTI part of CAN's core architecture?	No. The project's official Tech Stack (Project Design Documentation v1.1, Section 6) lists Keycloak, Sentinel/SIEM-equivalent, Tableau, Monday.com, Word/Excel, Lucidchart, and GitHub. OpenCTI does not appear in the project's designed tech stack
Why is it being addressed here at all?	It is named explicitly in the separate Cyber Threats and Vulnerabilities course rubric under "Implement Threat Intelligence Principles," which requires implementation of OpenCTI with at least 2 connectors. This is a course grading requirement layered on top of the CAN narrative, not a CAN architectural requirement
Current status	Instructor-confirmed: documentation of OpenCTI's role and integration plan satisfies this stage of the requirement. Live Docker deployment is documented below as planned future work, not yet implemented

How OpenCTI Would Integrate with Care Access Navigator (Planned)

If implemented, OpenCTI would not replace any existing CAN component — it would sit alongside the Monitoring segment (VLAN 40) as a proposed addition, formalizing threat data that is already documented for this project into a connected, queryable knowledge base:

CAN Artifact (already documented)	How It Would Map into OpenCTI
IOCs above (sample hash, exfil domain)	Entered as Indicator objects, each linked to the Malware object for AgentTesla
APT mapping (SilverTerrier, Orangethreat)	Entered as Intrusion-Set (threat actor) objects, linked to the Malware and Indicator objects above, and to Attack-Pattern objects for the MITRE ATT&CK; techniques observed
AgentTesla incident at HRMC	Entered as an Incident/Report object, linking the affected asset to the malware, actor, and indicator objects above — giving one clickable view of the whole event
Proposed connectors (2 required by rubric)	MITRE ATT&CK; connector — auto-imports the full Enterprise ATT&CK; framework so techniques link automatically; AlienVault OTX connector — pulls community-submitted IOCs, cross-referencing HRMC's own indicators against a live external feed

Value if implemented: This would turn CAN's threat data from static report tables into a connected system that mirrors how a real hospital SOC/GRC team tracks threats over time — supporting faster incident correlation and directly feeding prioritization decisions in the risk register.